



BLM3620 Digital Signal Processing*

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*Based on lecture notes from Ali Can Karaca & Ahmet Elbir

Lecture #1 – Introduction to DSP

- General Information about the Course
- Recommended Documents
- Introduction
- DSP Applications
- Basic Signal Operations

Important Materials:

- ✂️ • James H. McClellan, R. W. Schafer, M. A. Yoder, *DSP First Second Edition*, Pearson, 2015.
- Lizhe Tan, Jean Jiang, *Digital Signal Processing: Fundamentals and Applications*, Third Edition, Academic Press, 2019.

Auxiliary Materials:

- Prof. Sarp Ertürk, *Sayısal İşaret İşleme*, Birsen Yayınevi.
- Prof. Nizamettin Aydın, DSP Lecture Notes.
- J. G. Proakis, D. K. Manolakis, *Digital Signal Processing Fourth Edition*, Pearson, 2014.
- J. K. Perin, *Digital Signal Processing, Lecture Notes*, Stanford University, 2018.

General Information



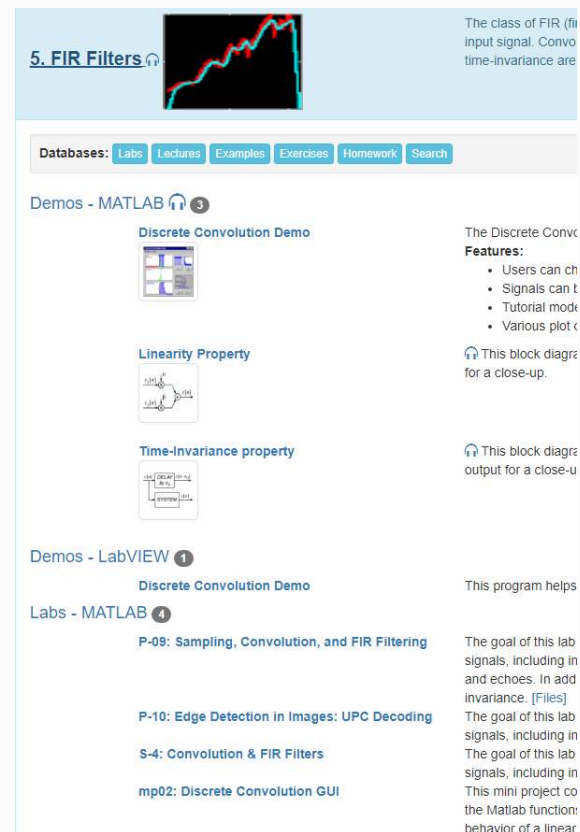
Ders Adı	Kodu	Yerel Kredi	AKTS	Ders (saat/hafta)	Uygulama (saat/hafta)	Laboratuvar (saat/hafta)
Sayısal İşaret İşleme	BLM3620	3	5	3	0	0

Değerlendirme Sistemi		
Etkinlikler	Sayı	Katkı Payı
Devam/Katılım		
Laboratuvar		
Uygulama		
Arazi Çalışması		
Derse Özgü Staj		
Küçük Sınavlar/Stüdyo Kritiği		
Ödev	4	30
Sunum/Jüri		
Projeler		
Seminer/Workshop		
Ara Sınavlar	1	30
Final	1	40
Dönem İçi Çalışmaların Başarı Notuna Katkısı		60
Final Sınavının Başarı Notuna Katkısı		40

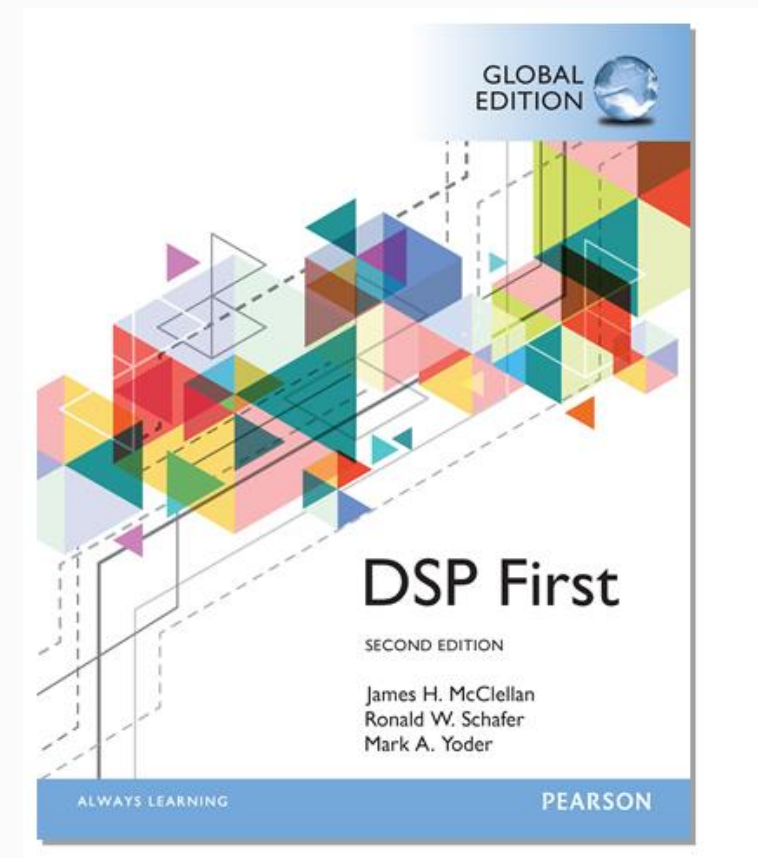
James H. McClellan, R. W. Schafer, M. A. Yoder, *DSP First Second Edition*, Pearson, 2015.

<https://dspfirst.gatech.edu/>

- Demos,
- Examples,
- Exercises,
- Lectures,
- Labs...



The screenshot shows the DSP First website interface. At the top, there's a header with the title "5. FIR Filters" and a small plot. Below this is a navigation bar with tabs for "Databases: Labs, Lectures, Examples, Exercises, Homework, Search". The main content area is divided into sections for "Demos - MATLAB", "Demos - LabVIEW", and "Labs - MATLAB". Each section lists various demos and labs with brief descriptions and icons. For example, under "Demos - MATLAB", there are "Discrete Convolution Demo", "Linearity Property", and "Time-Invariance property". Under "Labs - MATLAB", there are "P-09: Sampling, Convolution, and FIR Filtering", "P-10: Edge Detection in Images: UPC Decoding", "S-4: Convolution & FIR Filters", and "mp02: Discrete Convolution GUI".



W.	Date	Topics
1	23. Feb. 2024	Introduction to DSP and MATLAB
2	1. Mar. 2024	Sinuzoids and Complex Exponentials
3	8. Mar. 2024	Spectrum Representation
4	15. Mar. 2024	Sampling and Aliasing
5	22. Mar. 2024	Discrete Time Signal Properties and Convolution
6	29. Mar. 2024	Convolution and FIR Filters
7	5. Apr. 2024	Frequency Response of FIR Filters
8	12. Apr. 2024	Ramadan Feast
9	19. Apr. 2024	Midterm
10	26. Apr. 2024	Discrete Time / Discrete Fourier Transform and Properties
11	3. May. 2024	Fast Fourier Transform and Windowing
12	10. May. 2024	z- Transforms
13	17. May. 2024	FIR Filter Design and Applications
14	24. May. 2024	IIR Filter Design and Applications

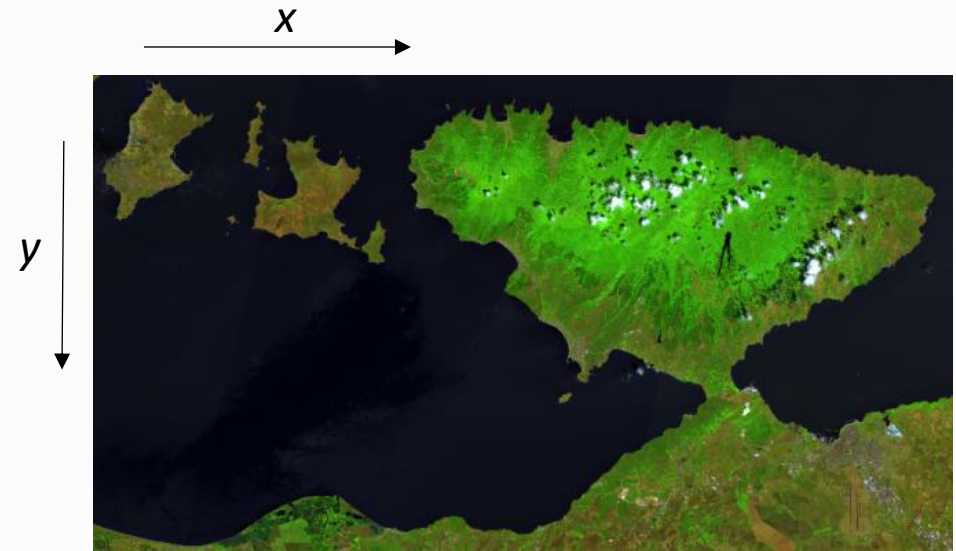
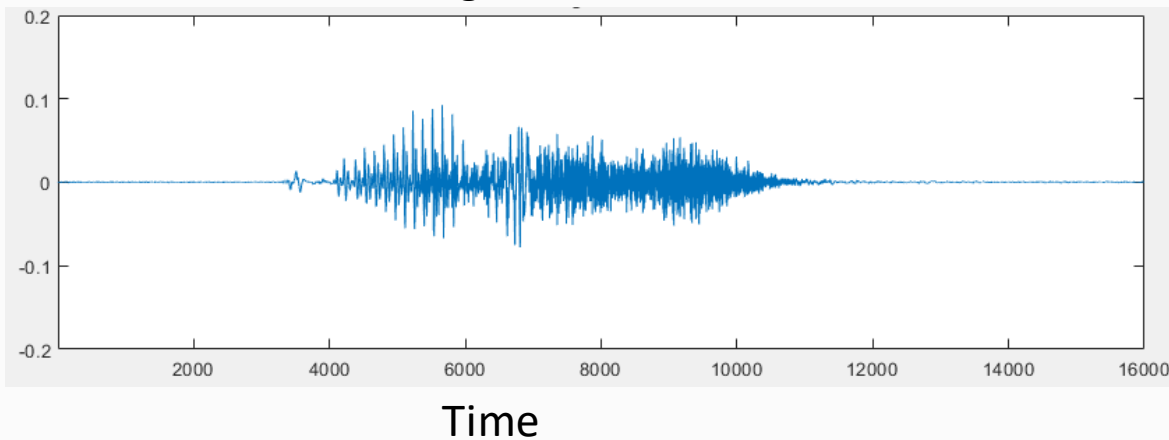
For more details -> Bologna page: <http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=5730&aid=3>

COURSE OBJECTIVE



- Students will be able to:
- Understand **mathematical** descriptions of signal processing **algorithms** and express those algorithms as computer **implementations** (MATLAB)
- What are your objectives?

«YES» Word Sound Signal



Mathematical Function: $I(x, y)$

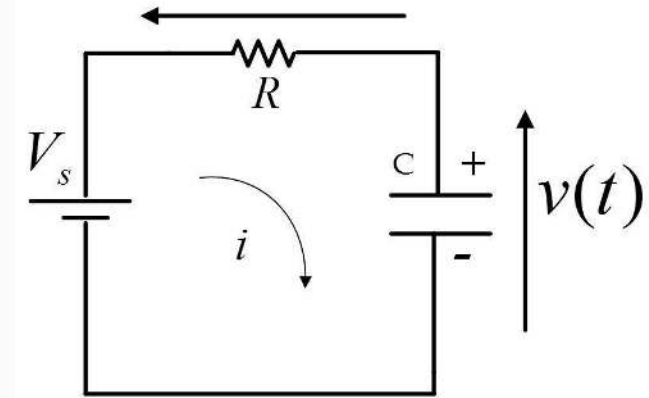
What are Signals and Systems?

► Signal:

- any physical quantity that varies with time, space, or any other independent variable or variables
- Examples: pressure as a function of altitude, sound as a function of time, color as a function of space, ...
- $x(t) = \cos(2\pi t)$, $x(t) = 4\sqrt{t} + t^3$, $x(m, n) = (m + n)^2$

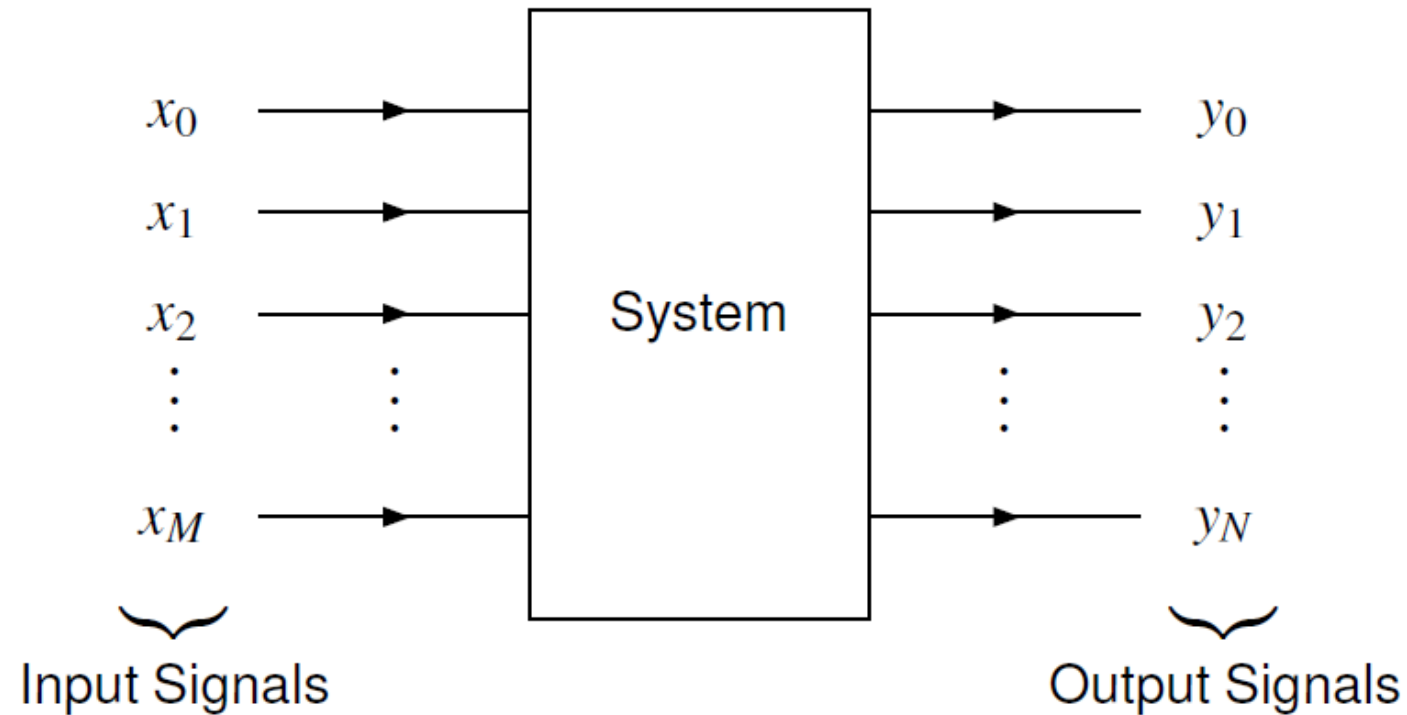
► System:

- a physical device that performs an operation on a signal
- Examples: analog amplifier, noise canceler, communication channel, transistor, ...
- $y(t) = -4x(t)$, $\frac{dy(t)}{dt} + 3y(t) = -\frac{dx(t)}{dt} + 6x(t)$,
 $y(n) - \frac{1}{2}y(n-2) = 3x(n) + x(n-2)$



What is a system?

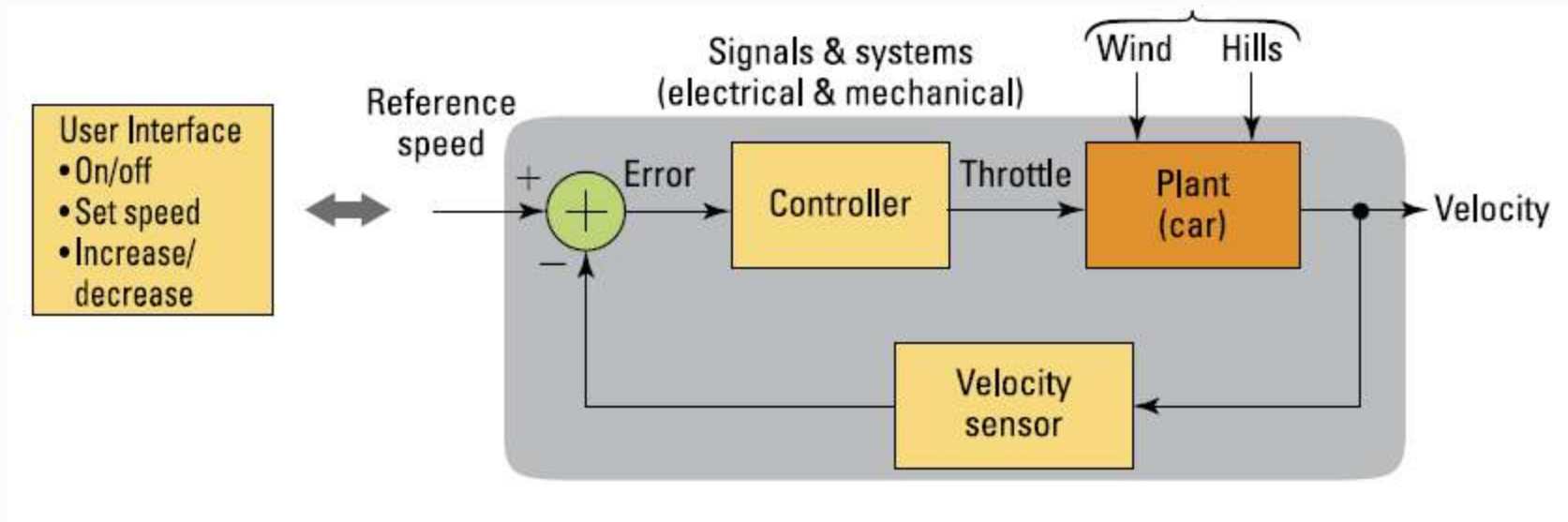
A **system** is an entity that processes one or more input signals in order to produce one or more output signals.



Example: Cruise Control System

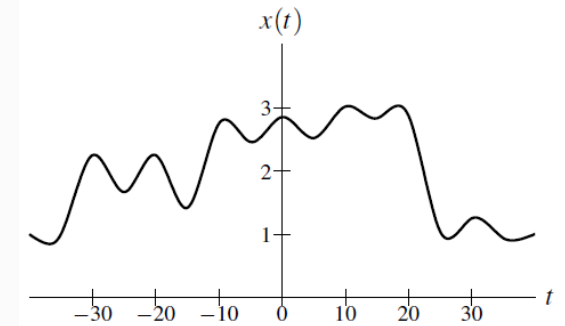
Input Signals: ?

Output Signals: Speed of car

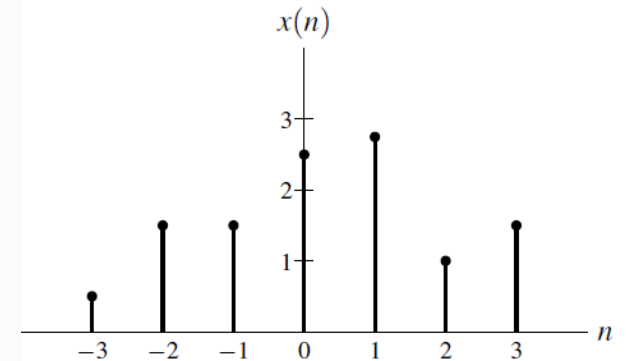


Classification of Signals

- Number of independent variables (i.e., dimensionality):
 - A signal with *one* independent variable is said to be **one dimensional** (e.g., audio).
 - A signal with *more than one* independent variable is said to be **multi-dimensional** (e.g., image).
- Continuous or discrete independent variables:
 - A signal with *continuous* independent variables is said to be **continuous time (CT)** (e.g., voltage waveform).
 - A signal with *discrete* independent variables is said to be **discrete time (DT)** (e.g., stock market index).



Continuous-Time (CT) Signal



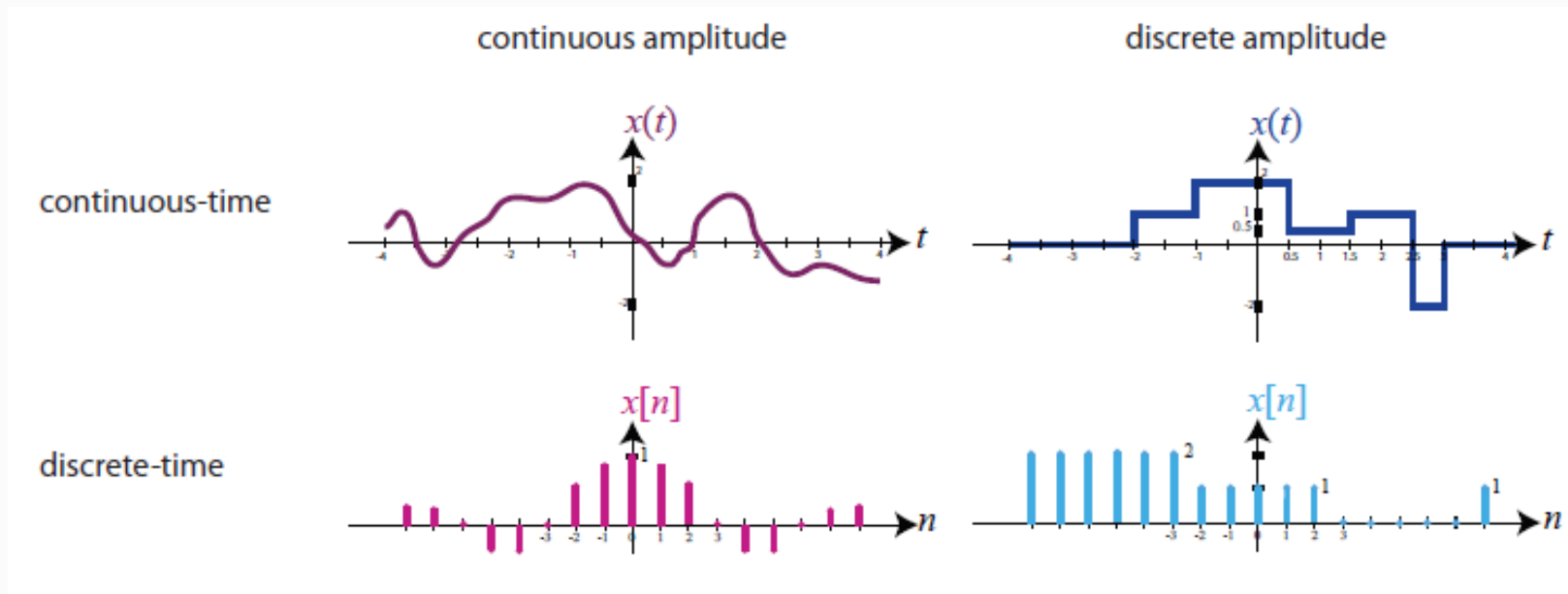
Discrete-Time (DT) Signal

Understanding Analog and Digital Signals



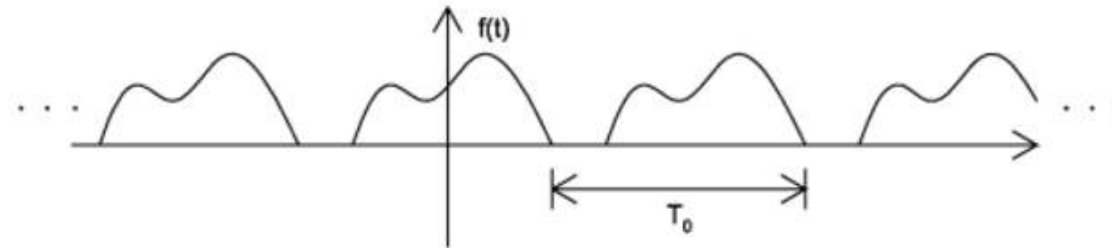
Analog signal-> continuous both in time and amplitude

Digital signal-> discrete both in time and amplitude



Classifications of Signals

Periodic vs. Aperiodic



3a: A periodic signal with period T_0



3b: An aperiodic signal

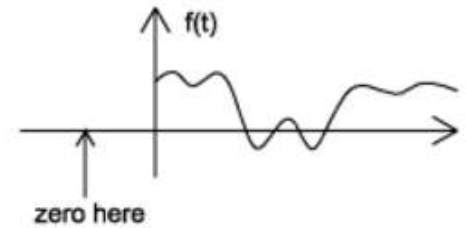
Periodic signals repeat with some **period** T , while aperiodic, or nonperiodic, signals do not. We can define a periodic function through the following mathematical expression, where t can be any number and T is a positive constant

$$f(t) = f(t + T)$$

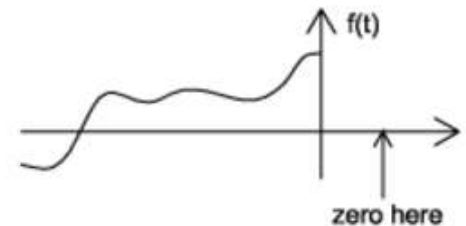
fundamental period of our function, $f(t)$, is the smallest value of T that the still allows Equation to be true.

Causal vs. Anticausal vs. Noncausal

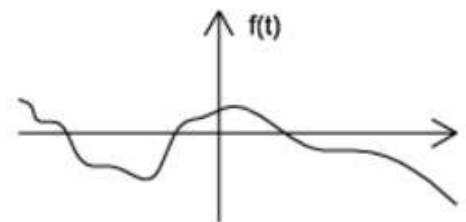
- **Causal** signals are signals that are zero for all negative time, while **anticausal** are signals that are zero for all positive time.
- **Noncausal** signals are signals that have nonzero values in both positive and negative time



4a: A causal signal



4b: An anticausal signal

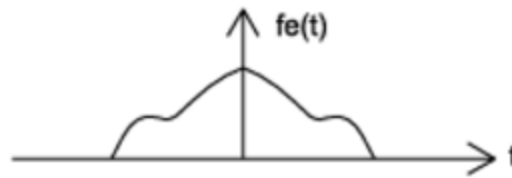


4c: A noncausal signal

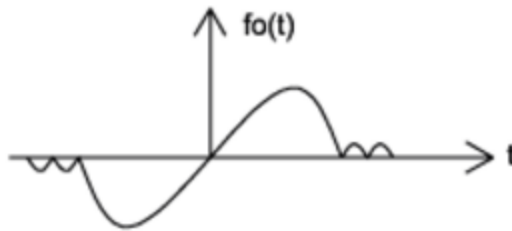
Classifications of Signals

- An **even signal** is any signal f such that $f(t)=f(-t)$. Even signals can be easily spotted as they are **symmetric** around the vertical axis.
- An **odd signal**, on the other hand, is a signal f such that $f(t)=-f(-t)$

Even vs. Odd

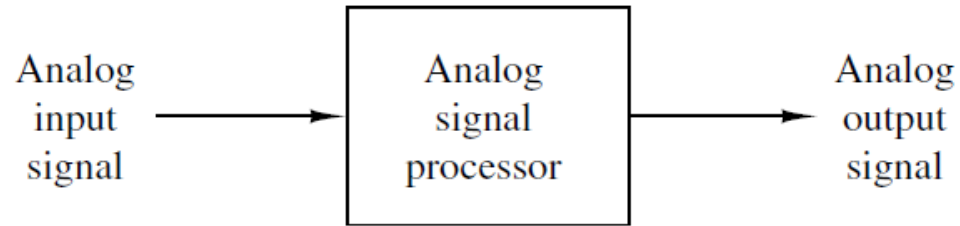


5a: An even signal



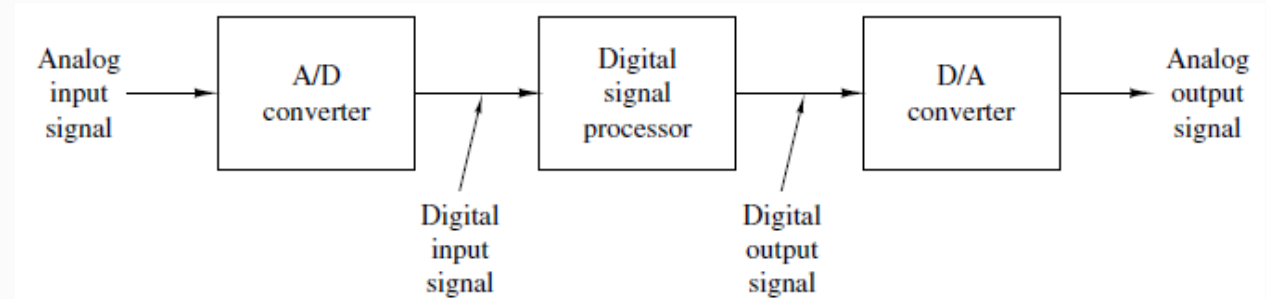
5b: An odd signal

Analog Systems vs. Digital Systems



Analog Systems:

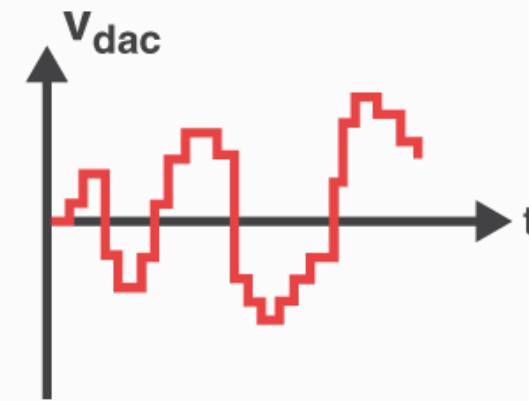
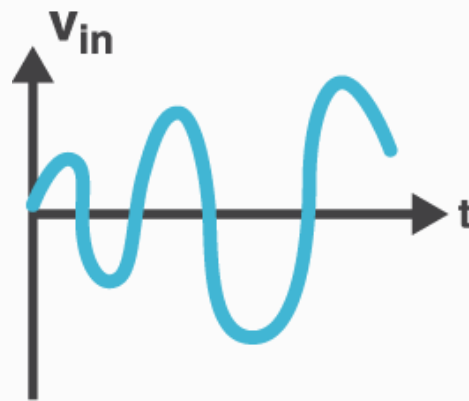
- *Directly use real-world signals.*
- *Do not need to an ADC or a DAC.*
- *Give the fastest application results.*



Digital Systems:

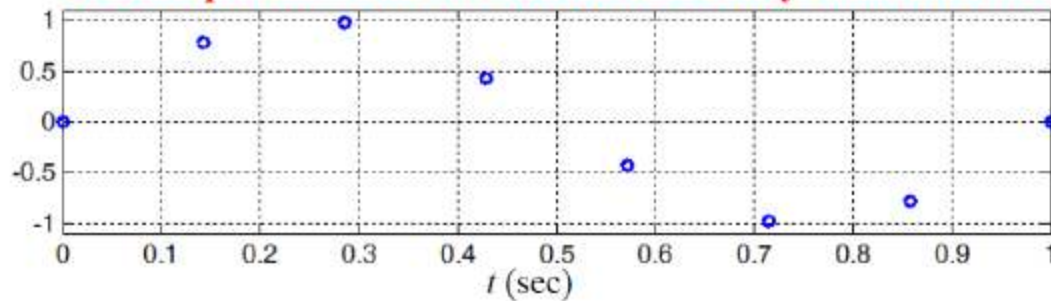
- *have lower distortions thanks to error correction.*
- *Digital signals can easily be compressed and saved.*
- *Do not include any R, L, C elements. (programmed on software)*
- *Are more stable and robust to the environmental conditions.*
- *Can be ported different hardware*

General Concept in DSP Applications

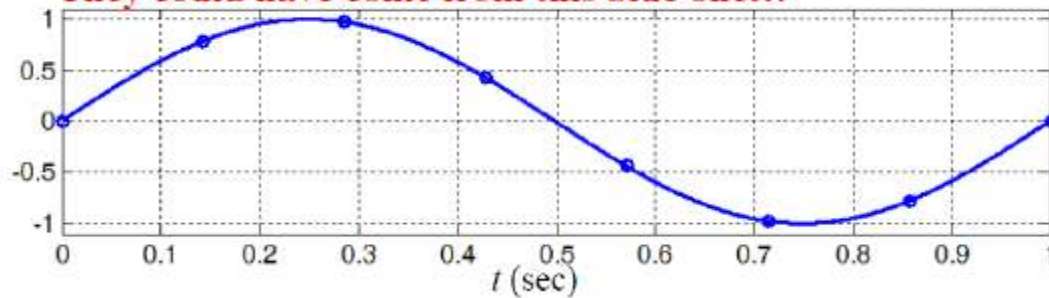


Sampling is very important !

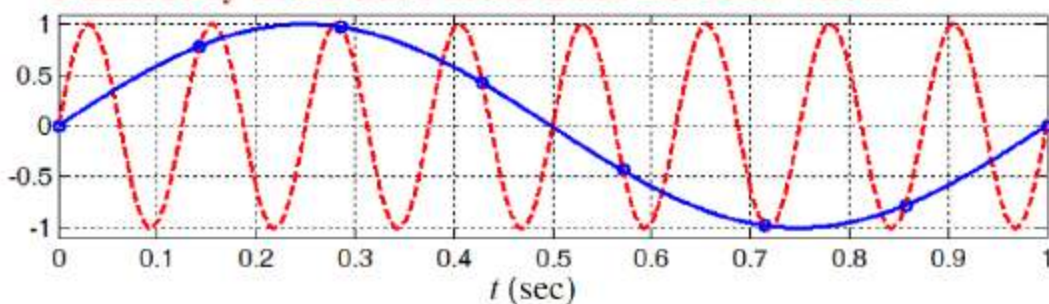
DT Samples.... What CT Sinusoid did they come from????



They could have come from this blue one...



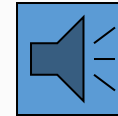
But...They could have come from this RED one!!!



Thus... if we want to be able to tell these two apart we need to sample faster!!

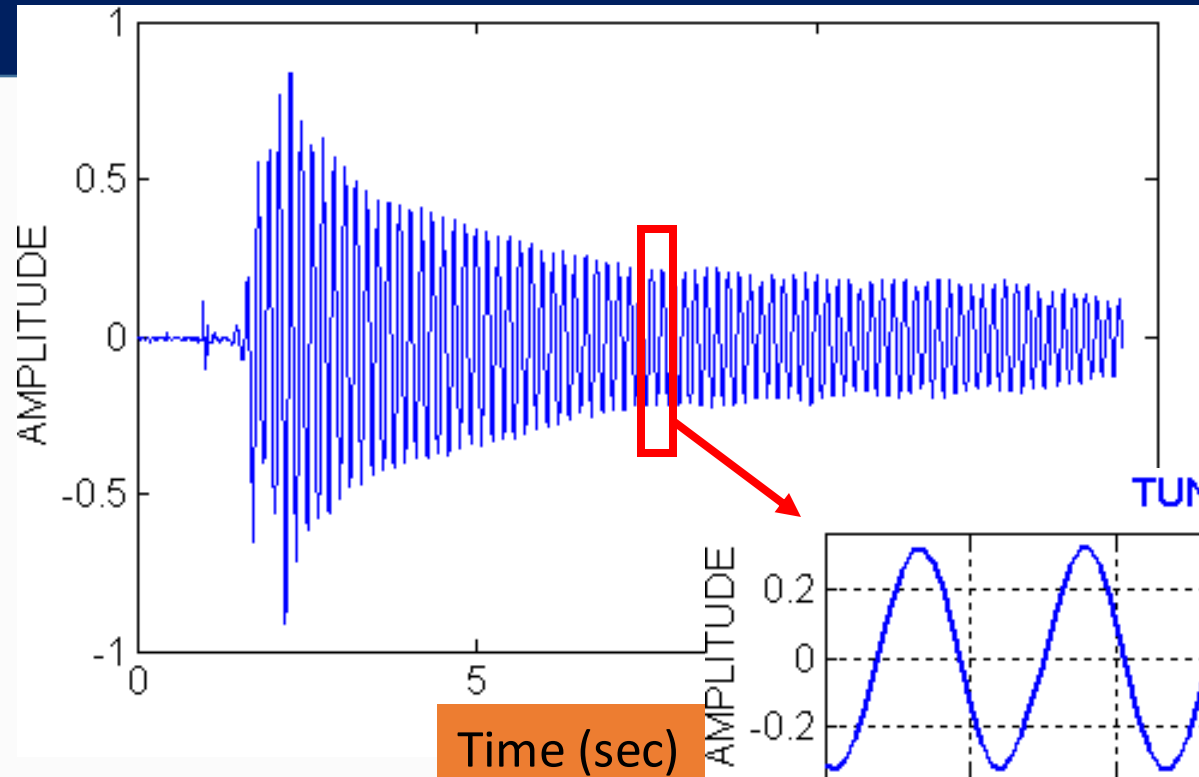
TUNING FORK EXAMPLE

- CD-ROM demo
- “A” is at 440 Hertz (Hz)
- Waveform is a SINUSOIDAL SIGNAL
- Computer plot looks like a sine wave
- This should be the mathematical formula:



$$A \cos(2\pi(440)t + \varphi)$$

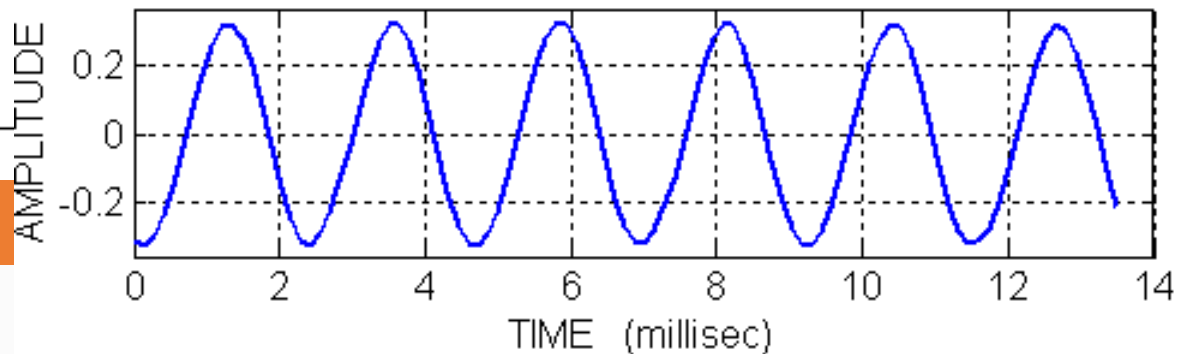
Tuning Fork: A-440 Waveform



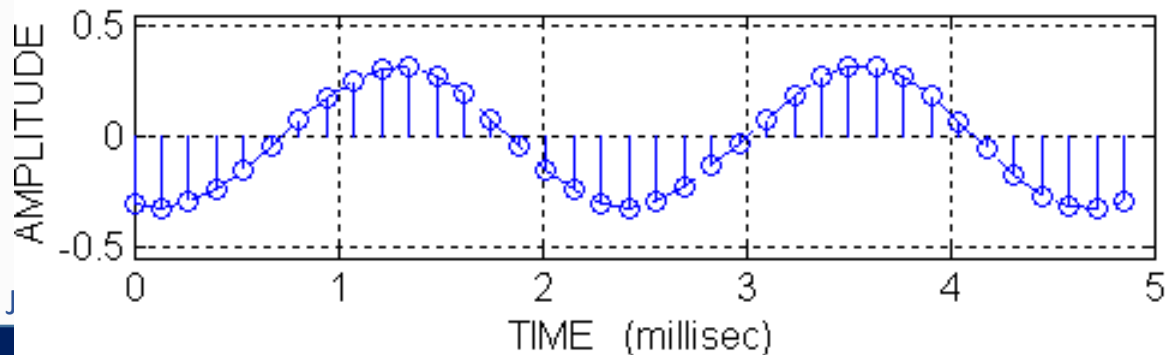
$$T \approx 8.15 - 5.85$$

$$= 2.3 \text{ ms}$$

TUNING FORK A-440



ZOOM in on TWO PERIODS



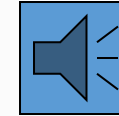
$$f = 1/T$$

$$= 1000 / 2.3$$

$$\approx 435 \text{ Hz}$$

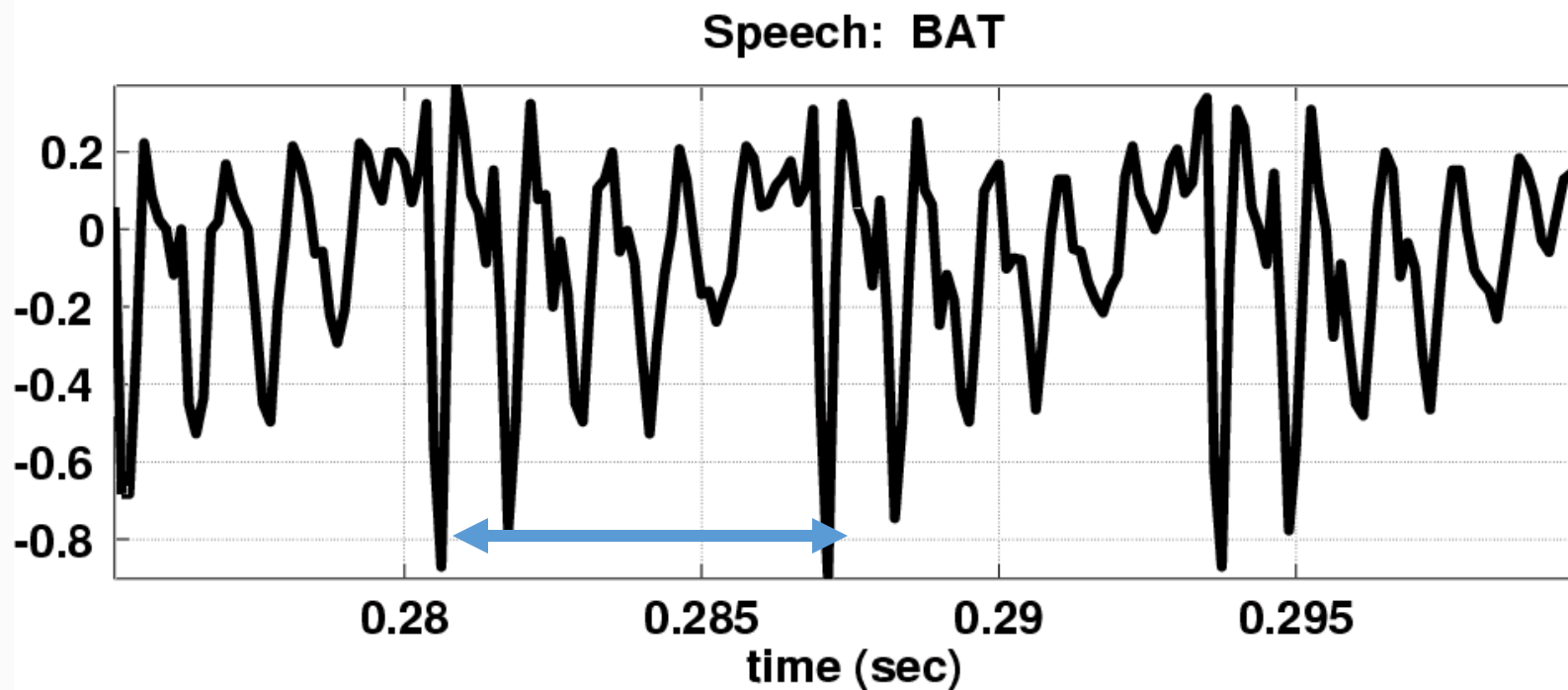
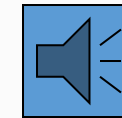
What about speech?

- More complicated signal (BAT.WAV)
- Waveform $x(t)$ is NOT a Sinusoid
- Theory will tell us
 - $x(t)$ is approximately a sum of sinusoids
 - FOURIER ANALYSIS
 - Break $x(t)$ into its sinusoidal components
 - Called the FREQUENCY SPECTRUM



What about speech?

- Nearly **Periodic** in Vowel Region
 - Period is (Approximately) $T = 0.0065$ sec

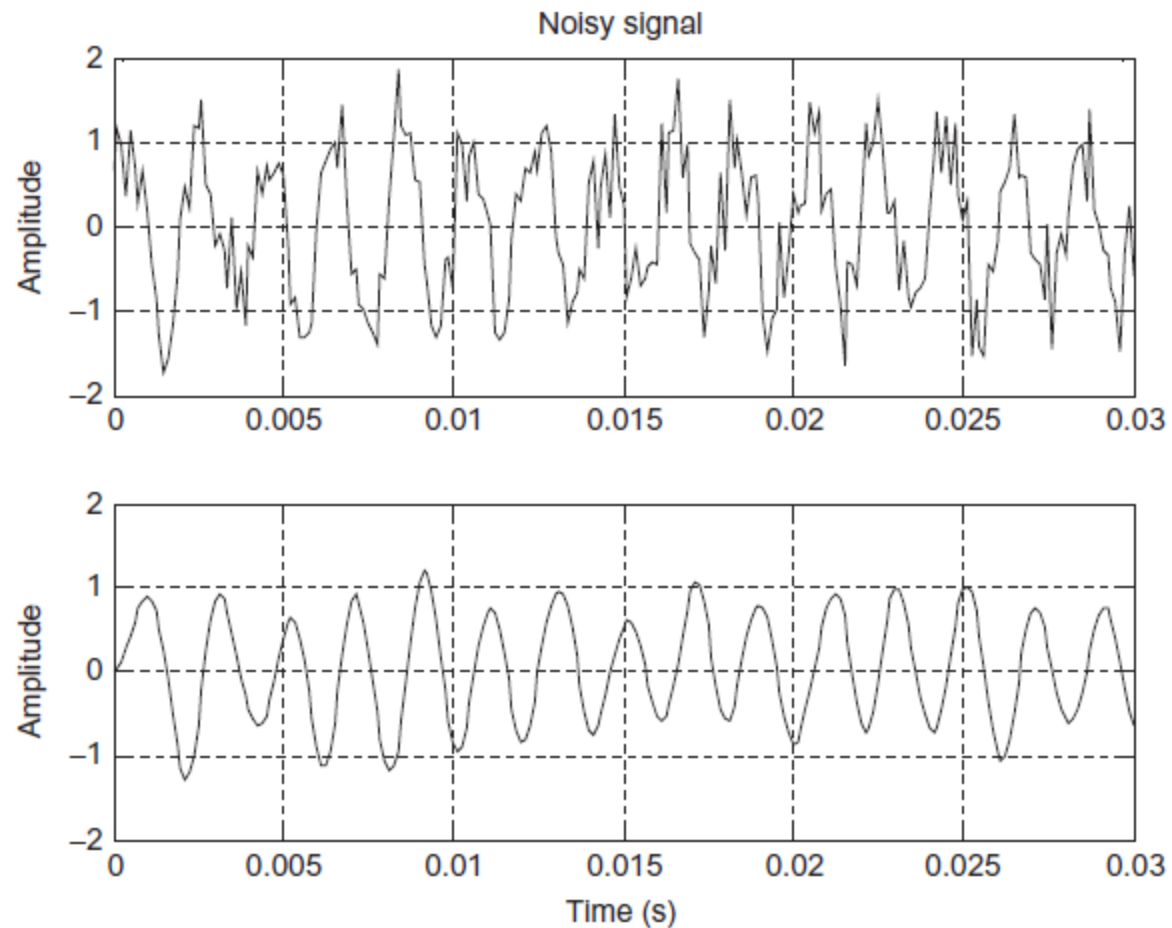


STORING DIGITAL SOUND



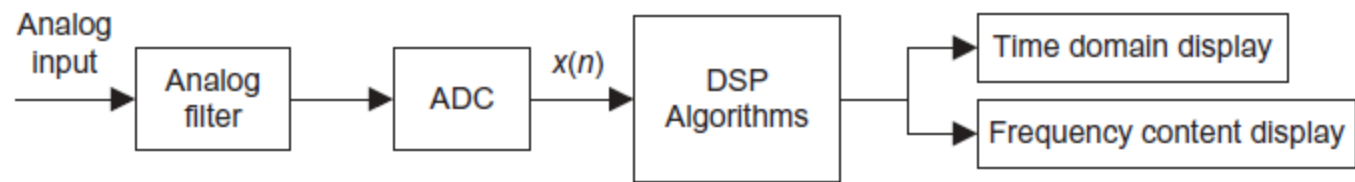
- $x[n]$ is a SAMPLED SINUSOID
 - A list of numbers stored in memory
- CD rate is 44,100 samples per second
- 16-bit samples
- Stereo uses 2 channels
- Number of bytes for 1 minute is
 - $2 \times (16/8) \times 60 \times 44100 = 10.584$ Mbytes

DSP Applications: Digital Filtering

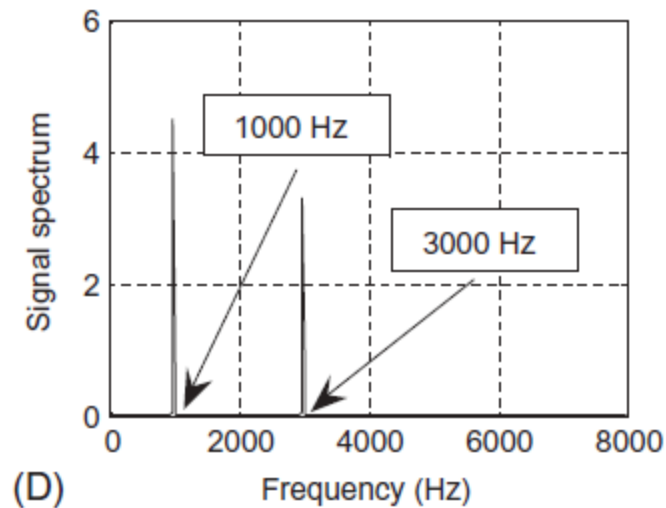
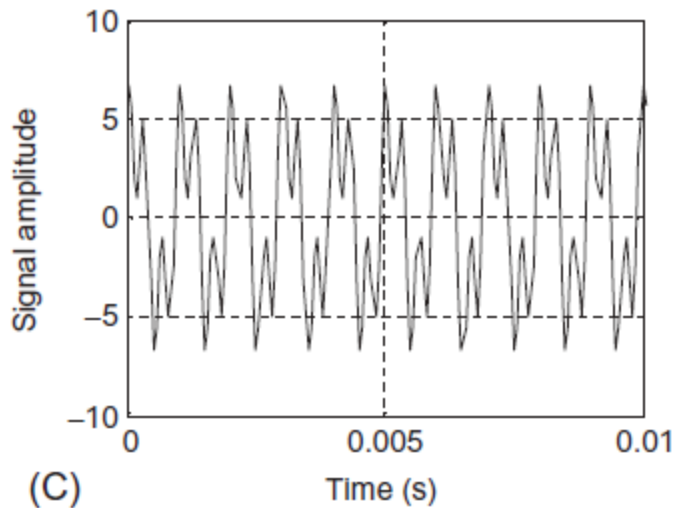
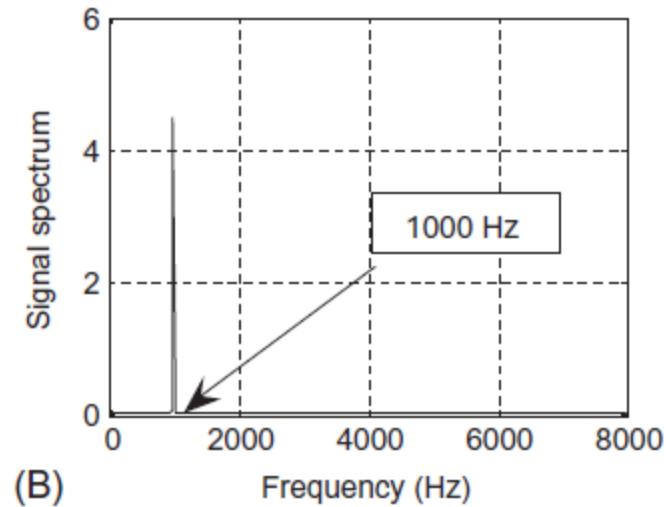
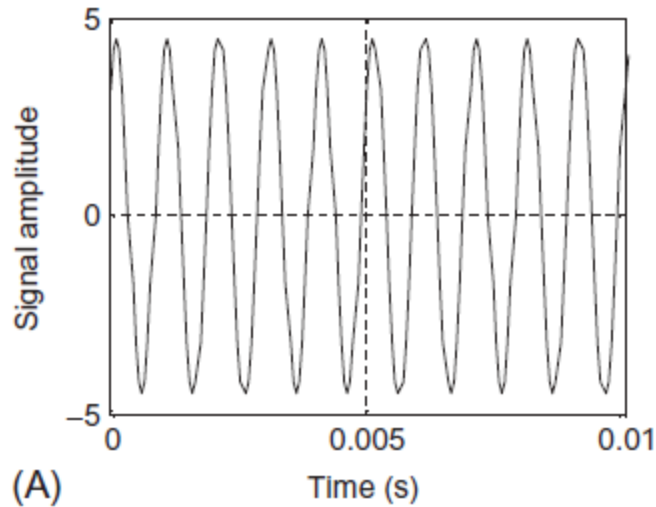


Which filter type should we use?

DSP Applications: Dig



We should go to frequency domain...



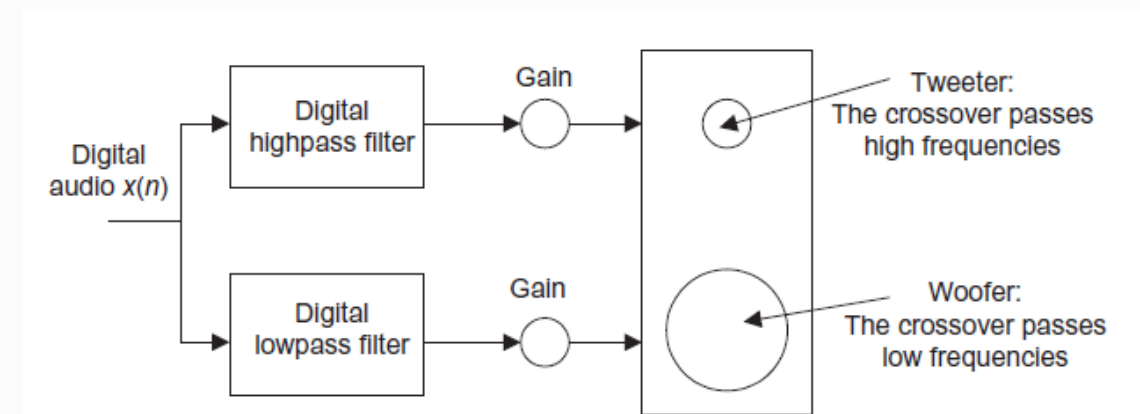
Noise Cancelling

Ref. J. Jiang and L. Tan

Some Real-World Apps: Digital Crossover Audio System

The incoming digital audio signal is split into two bands using a digital lowpass filter and a digital highpass filter in parallel. Then the separated audio signals are amplified. Finally, they are sent to their corresponding speaker drivers.

Although the traditional crossover systems are designed using the analog circuits, the digital crossover system offers a cost-effective solution with programmable ability, flexibility, and high quality.

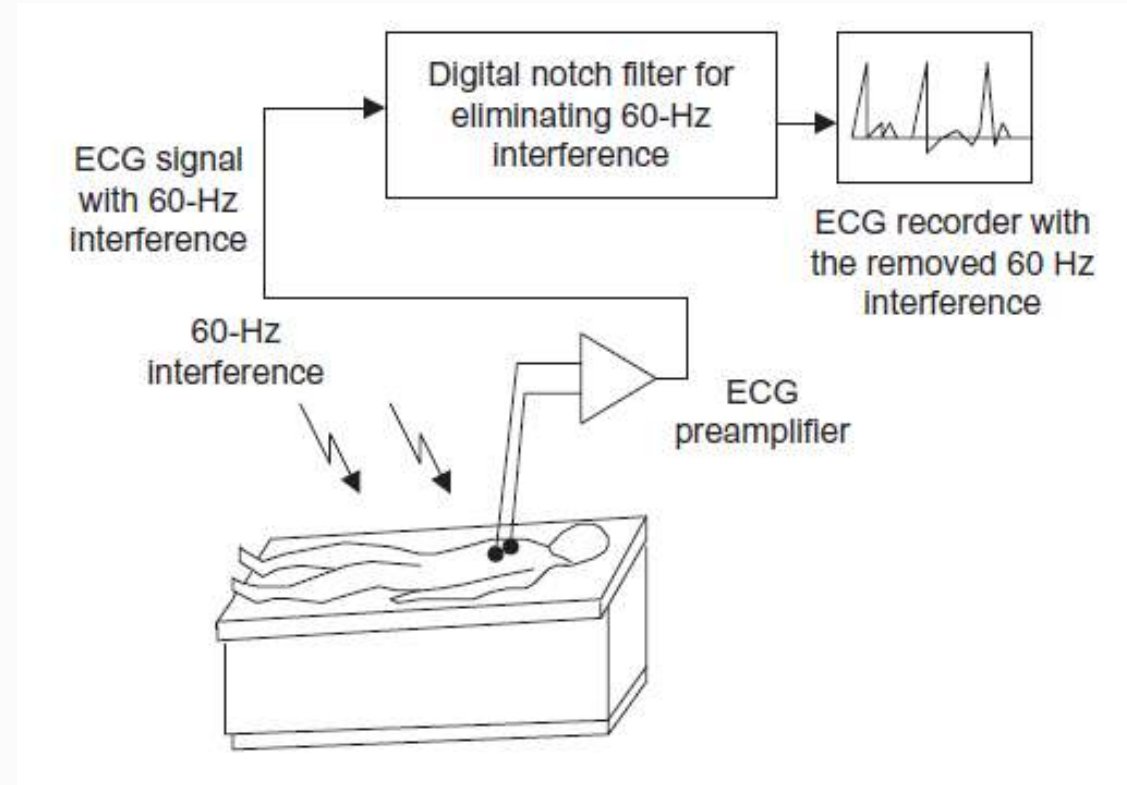


Some Real-World Apps: Filtering ECG Signal

Design a notch filter and apply it to digital signal

kalp ritminin genliği çok düşük, bunu preamplifier ile yükseltiyoruz. Ancak ortamdaki floresanlar 60 Hz de bir gürültü oluşturuyor (Bu olaya interference denir.). Bunu engellemek için 60 Hz deti bir sinyal yok eden, baskıncu bir notch filtreler kullanılıyor.

Radio frekansları ile teliz iletişimini bozma uygulaması.



Some Real-World Apps: Vibration Signature for Defect Det.

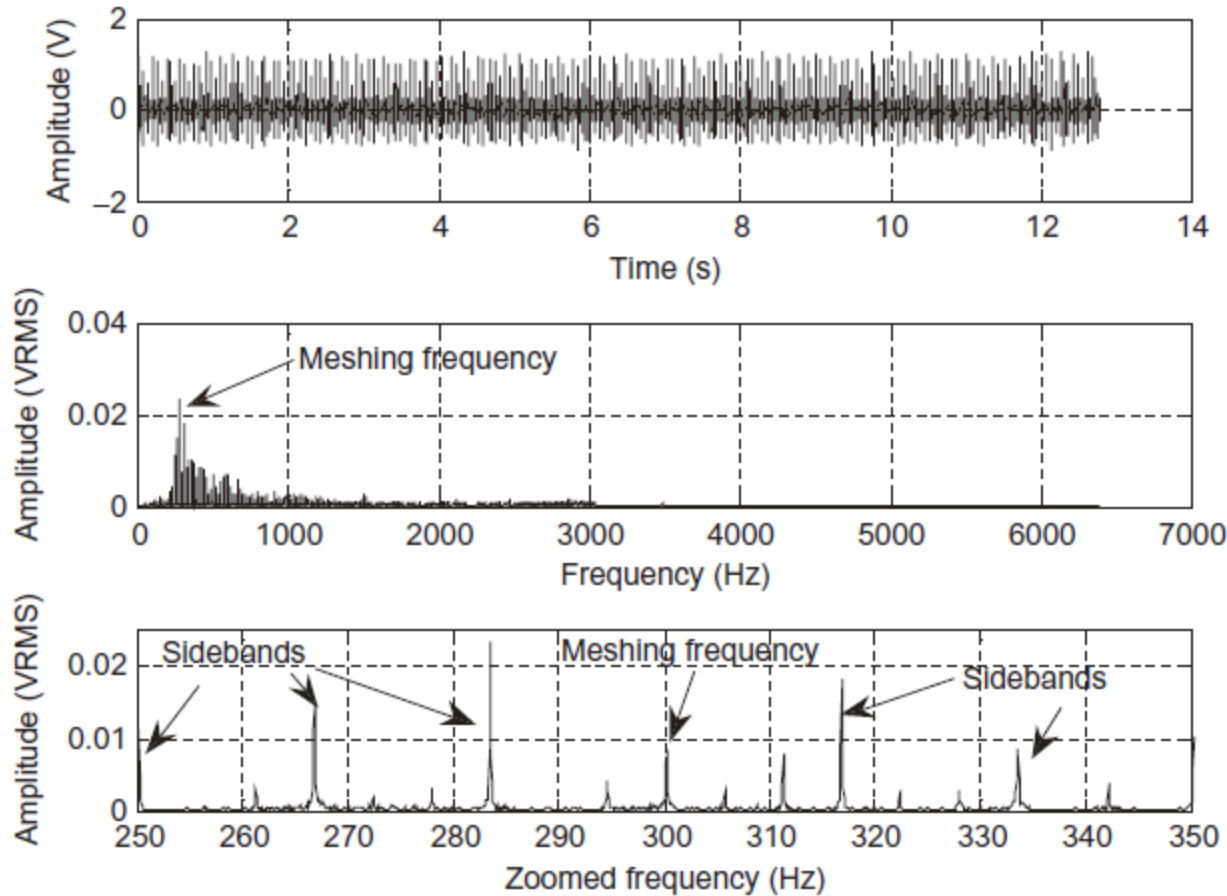
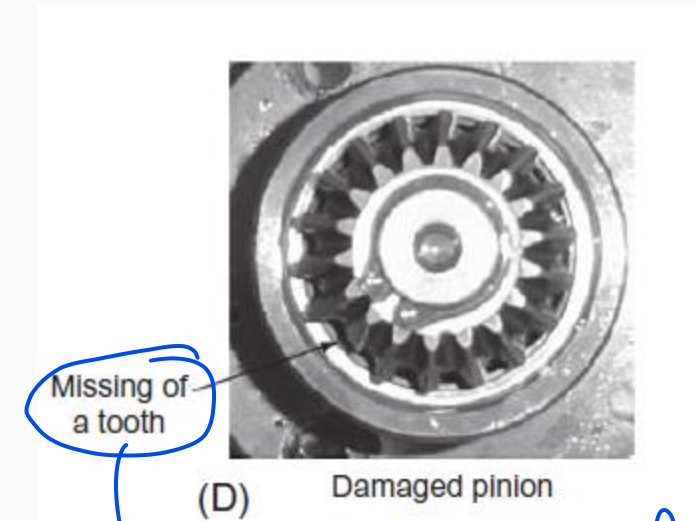


FIG. 1.13

Vibration signal and spectrum from the damaged gearbox.



g düşük olmaması için bir frekansı
ta ses çıkarır. Böylece bozuk olup
olmayan yer bulunur.

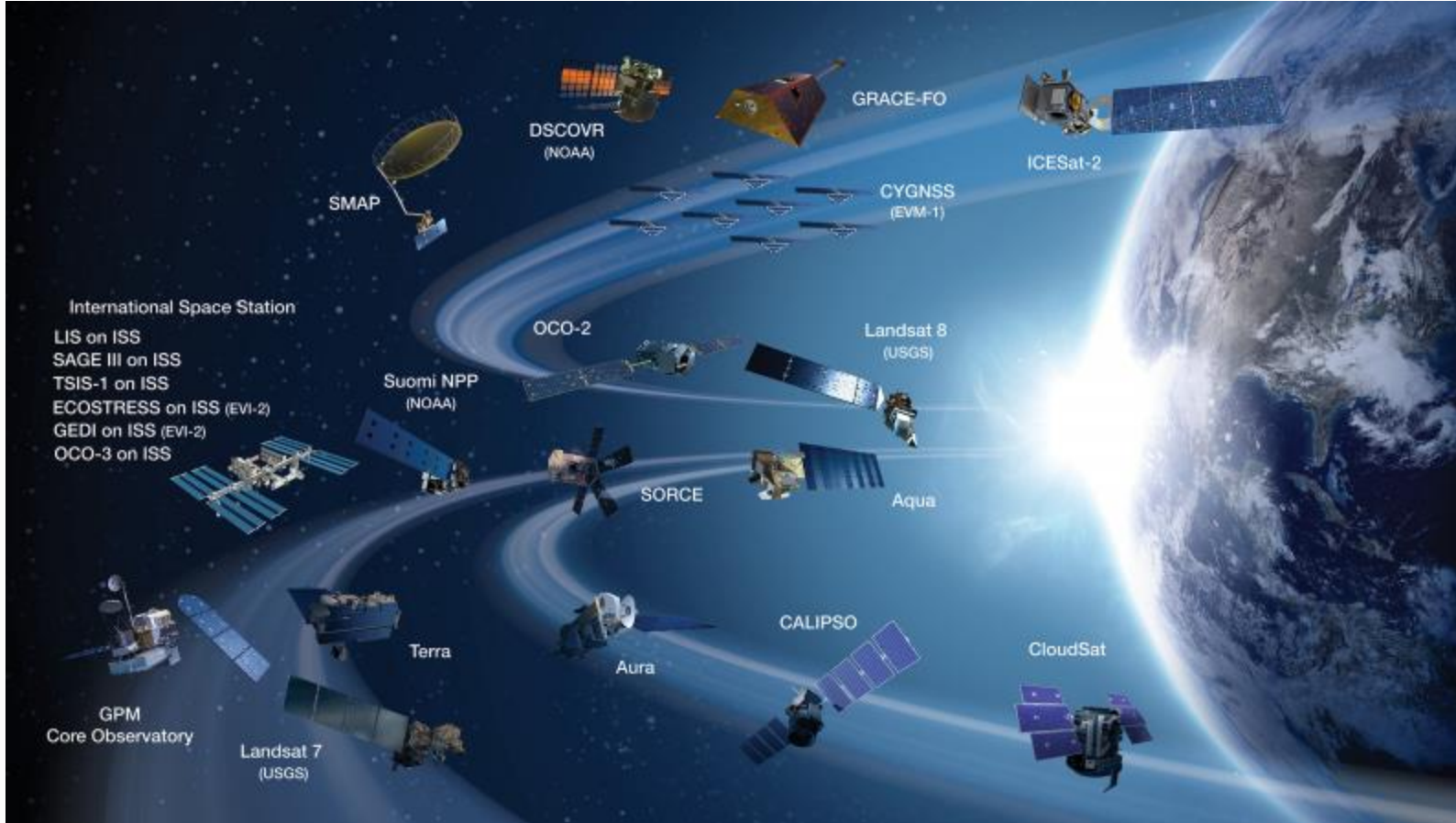
Other Applications

DSP = Swiss-Army-Knife of modern CEng

Alçı fedmi

- Communications (wireless, internet, GPS),
- Control and monitoring (cars, machines...),
- Multimedia (videos, cameras, ...),
- Healthcare (medical devices),
- More...

Satellite Image Compression



büyük verileri sıkıştırarak Dünya'ya gönderme de yine DSP kullanılır. Çok büyük boyutlu resimler.

Speech Spoofing
(konuşmanın zehirlenmesi)

Aptatuna da bir ses donunu büyük bir ses tonuna atma yapılabiliyor. Bir insanın sesi yerine başka bir insanın sesi kullanılabilir. (Güvenli fre kullanılıyor.)

HDR (High Dynamic Range)

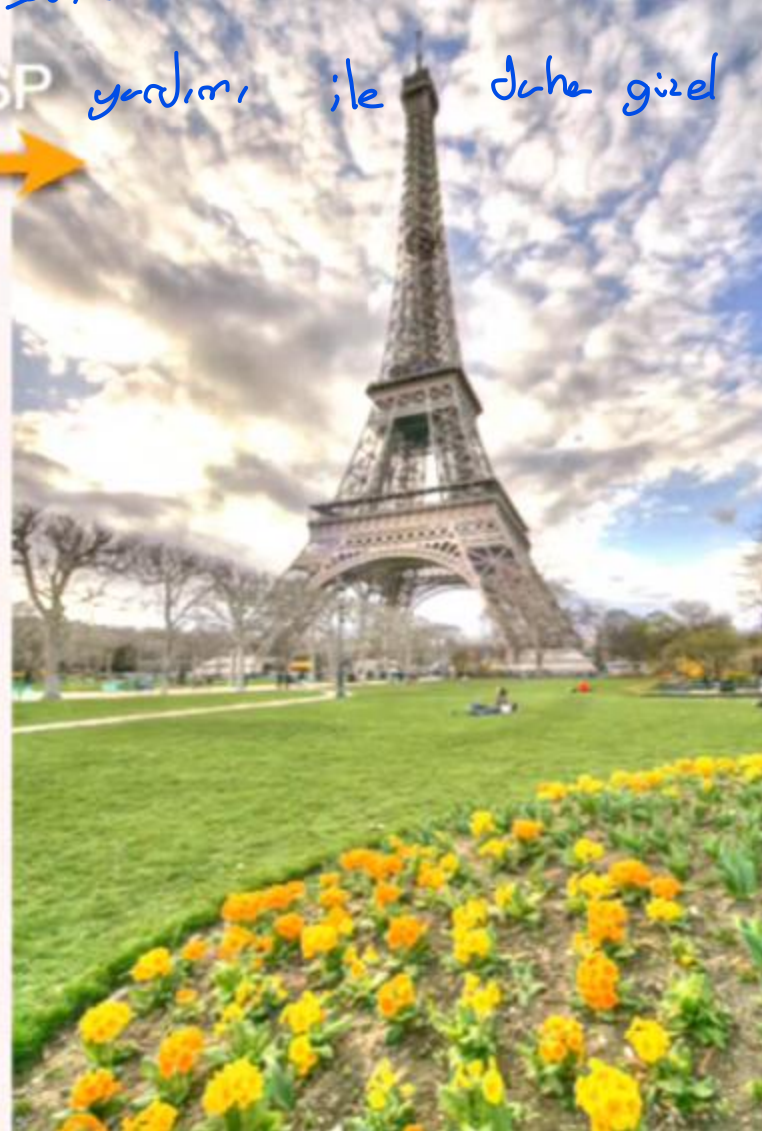
bulutlar yok

karanlık

detaylar yok

DSP

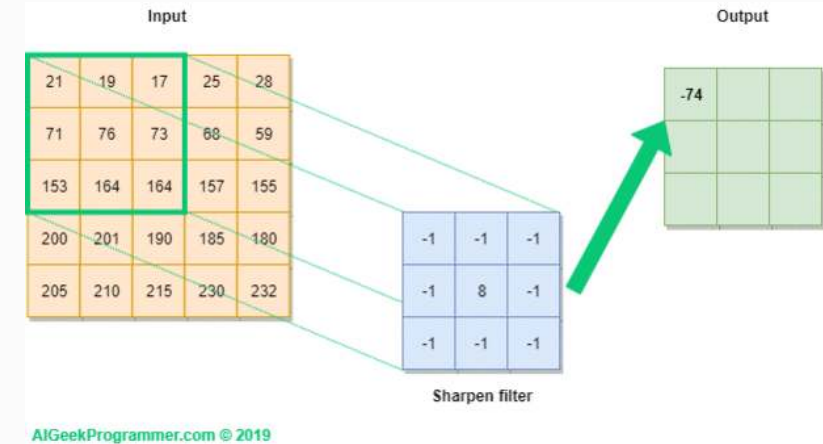
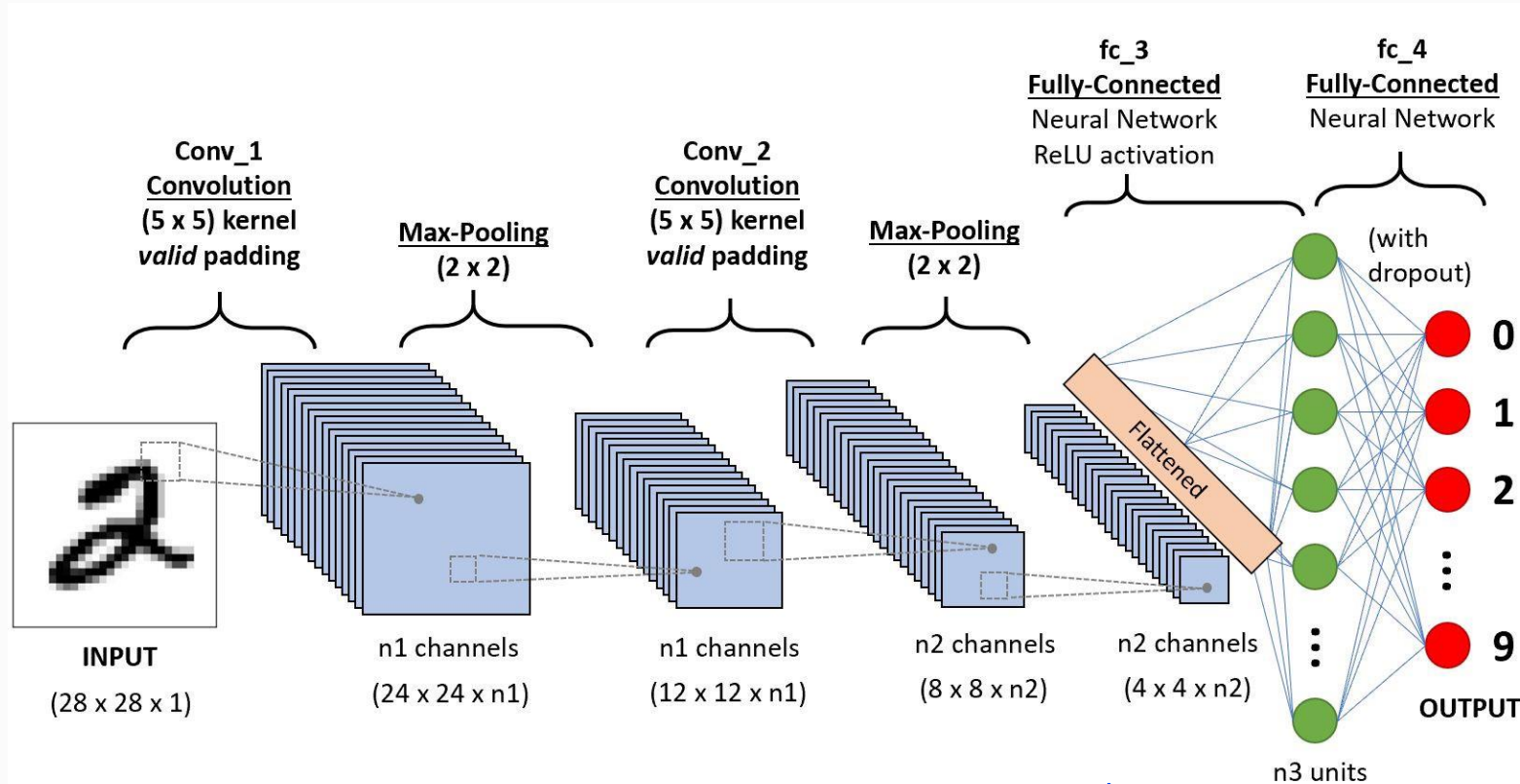
yardımı ile daha güzel bir foto



Now implemented in smart phones (HDR)

[*www.hdrsoft.com](http://www.hdrsoft.com)

Some Real-World Apps: Convolutional NNs



yazık sayının 2 olduğunu tahmin etmek için AI eğitimi. konvolüsyon kullanılıyor.

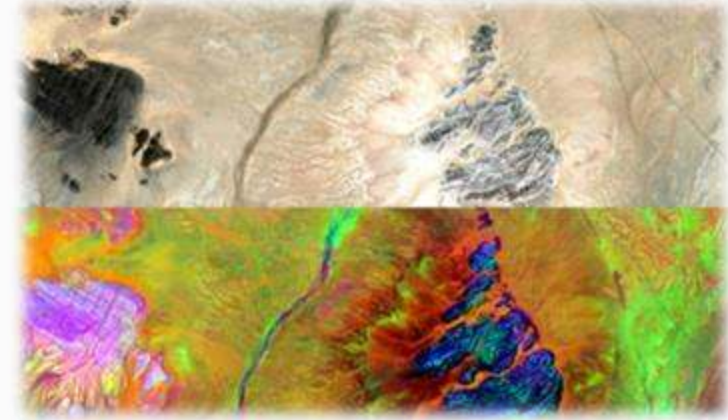
Image Processing – Remote Sensing



Change detection



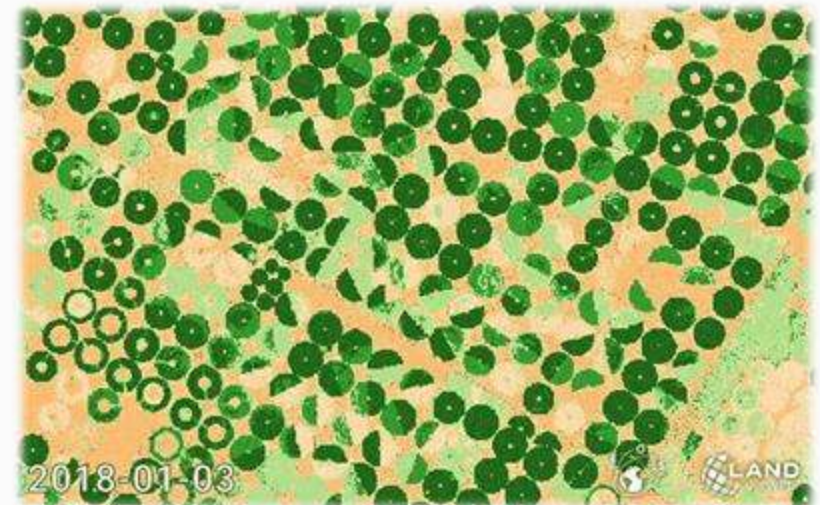
3D mapping



Mineral mapping

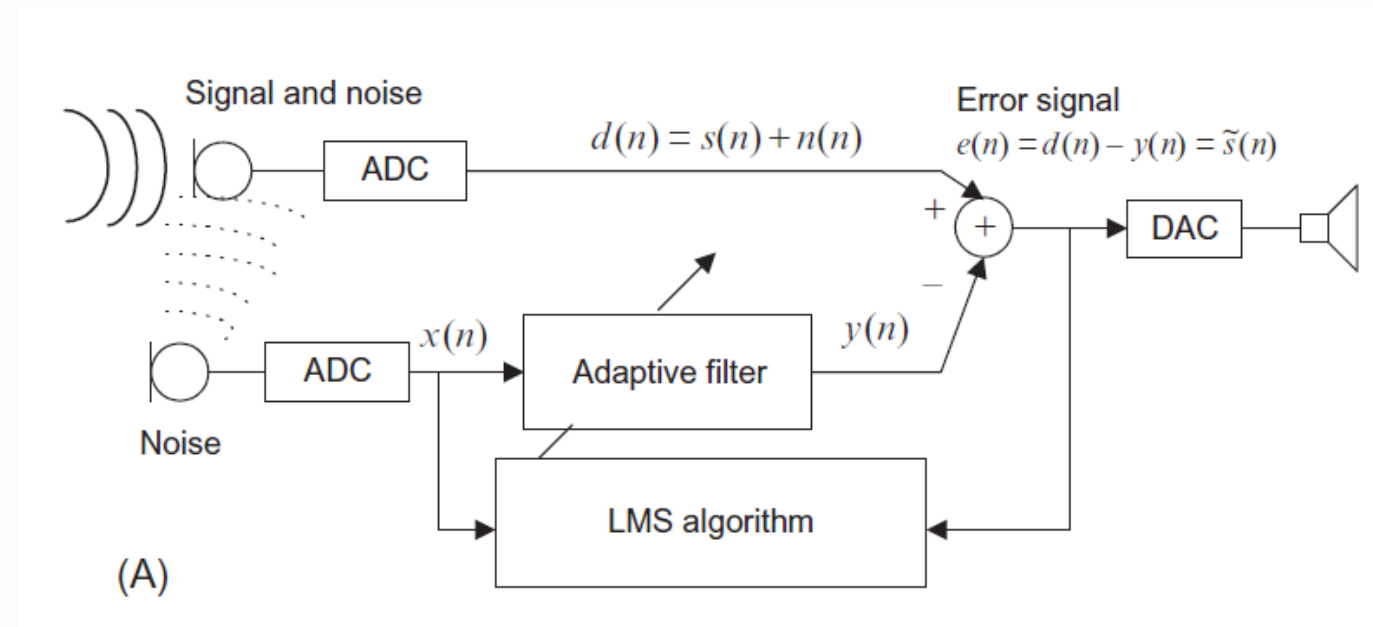


Target detection



agriculture

Some Real-World Apps: Noise Cancellation

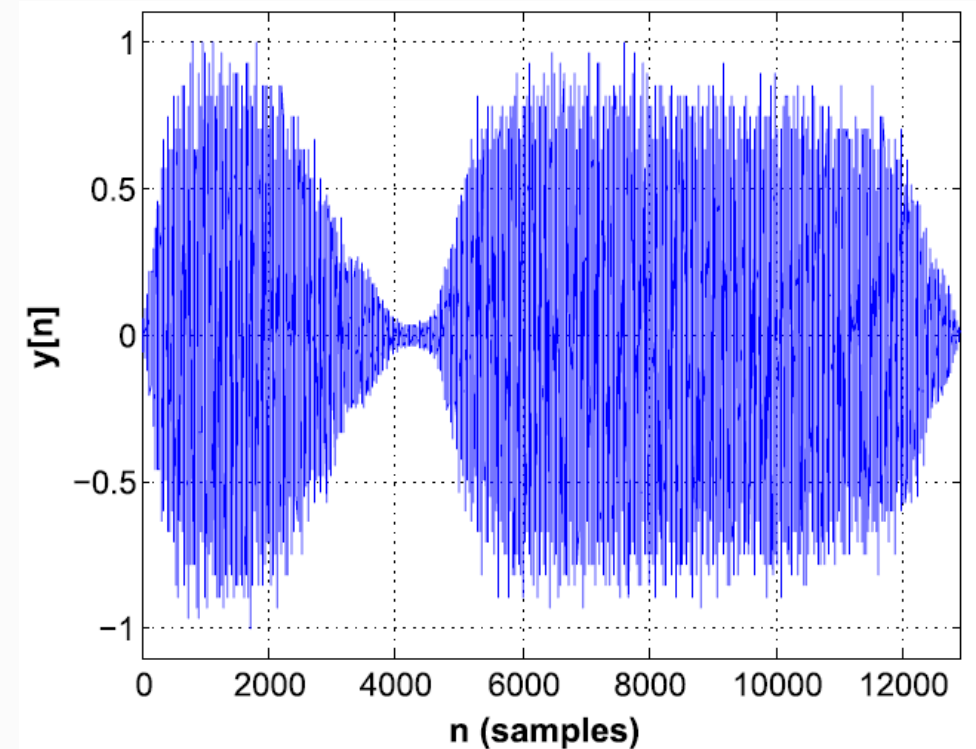


Let's take a look at train signal...

```
clear all;
load train;

%% Example---Listening to/plotting train signal
sound(y,Fs)
t=0:1/Fs:(length(y)-1)/Fs;
figure(2); plot(t,y'); grid
ylabel('y[n]'); xlabel('n')

%% Example---Using stem to plot 200 samples of train
figure(3)
n=100:299;
stem(n,y(100:299)); ylabel('y[n]'); xlabel('n')
title('Segment of train signal')
axis([100 299 -0.5 0.5])
```

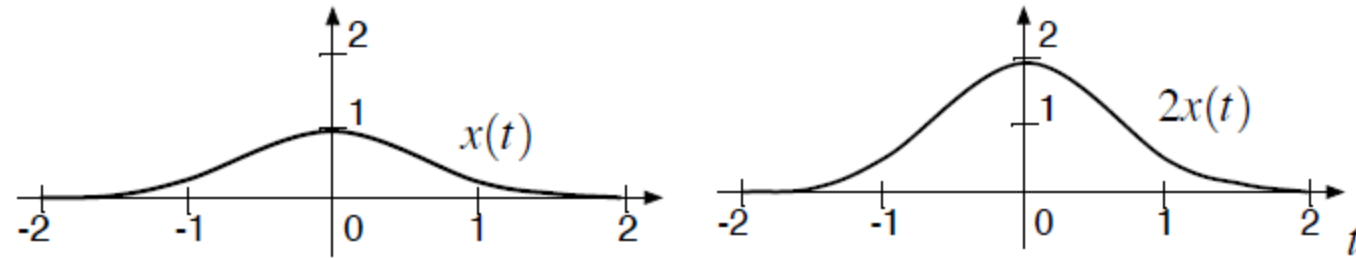


How can we decrease the amplitude of the sound??

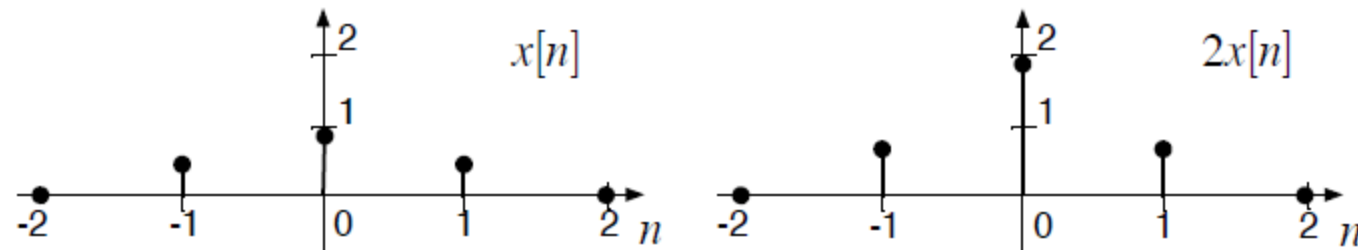
What about noisy signals??

Amplitude Scaling

- The scaled signal $ax(t)$ is $x(t)$ multiplied by the constant a



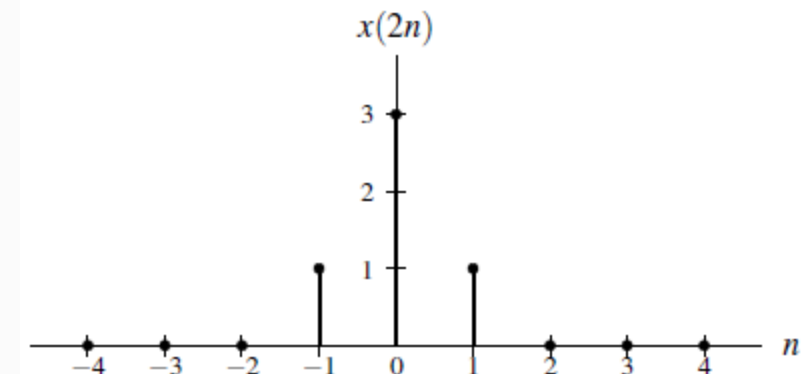
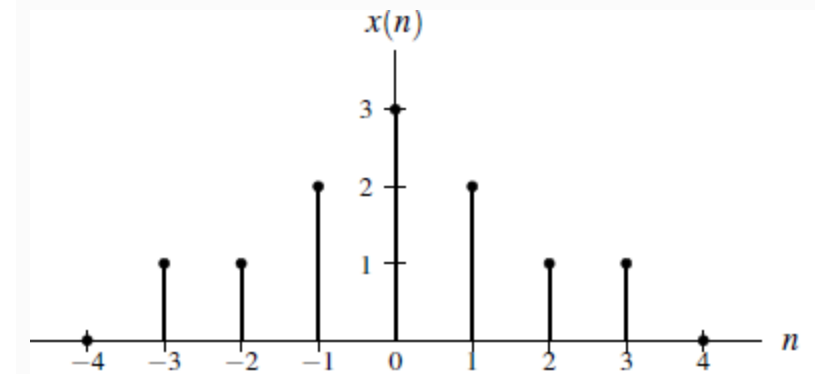
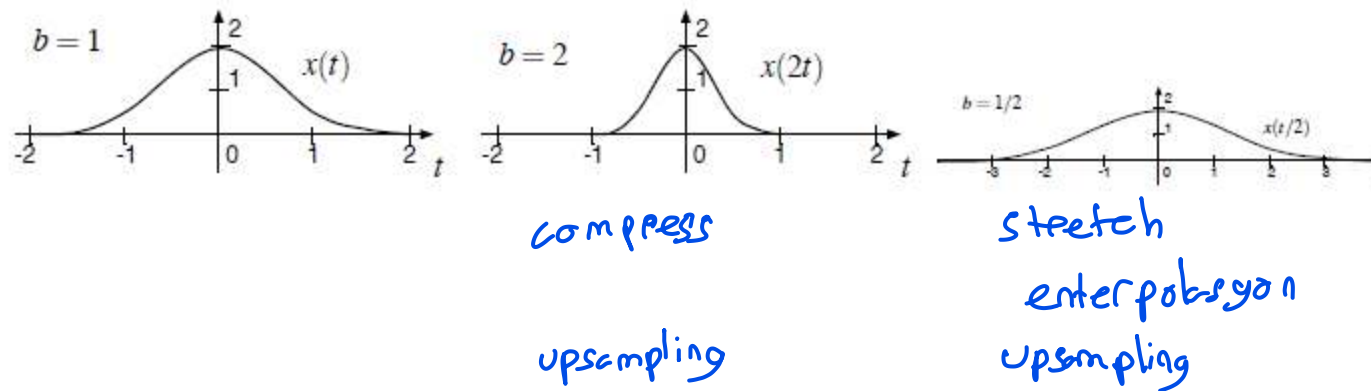
- The scaled signal $ax[n]$ is $x[n]$ multiplied by the constant a



Operations on Signals

Time Scaling, Continuous Time

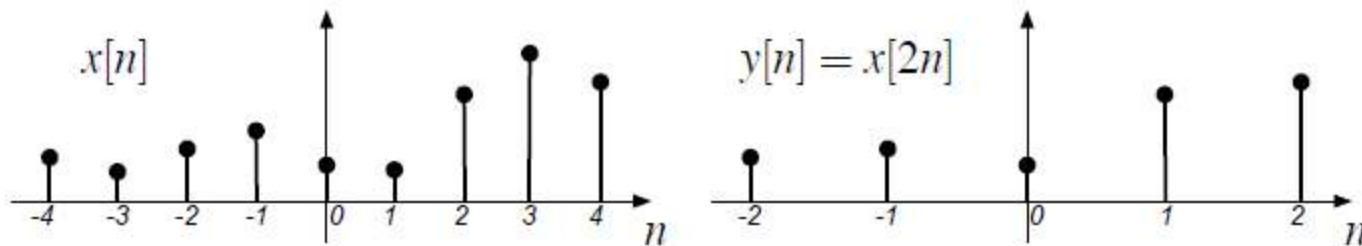
A signal $x(t)$ is scaled in time by multiplying the time variable by a positive constant b , to produce $x(bt)$. A positive factor of b either expands ($0 < b < 1$) or compresses ($b > 1$) the signal in time.



Time Scaling, Discrete Time

The discrete-time sequence $x[n]$ is *compressed* in time by multiplying the index n by an integer k , to produce the time-scaled sequence $x[nk]$.

- This extracts every k^{th} sample of $x[n]$.
- Intermediate samples are lost.
- The sequence is shorter.



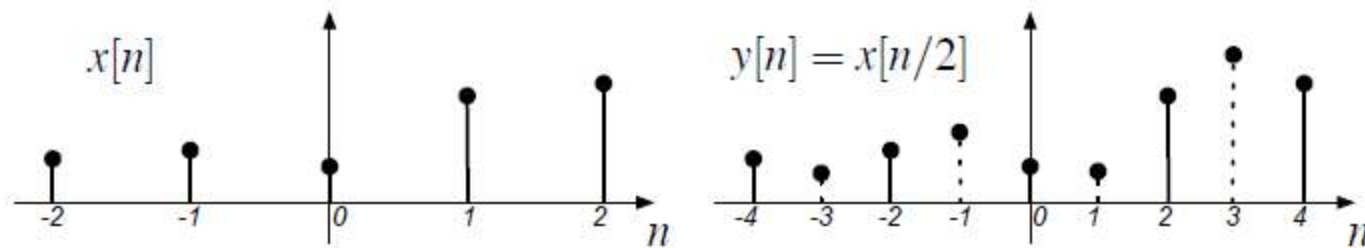
Called *downsampling*, or *decimation*.

Operations on Signals



The discrete-time sequence $x[n]$ is *expanded* in time by dividing the index n by an integer m , to produce the time-scaled sequence $x[n/m]$.

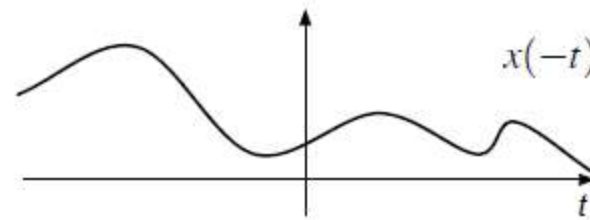
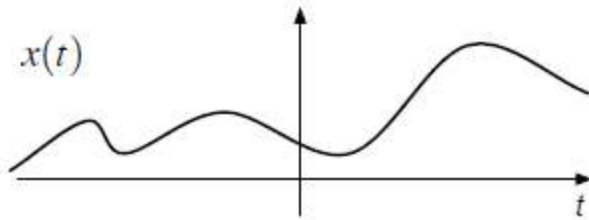
- This specifies every m^{th} sample.
- The intermediate samples must be synthesized (set to zero, or interpolated).
- The sequence is longer.



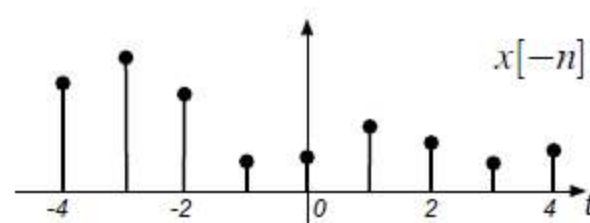
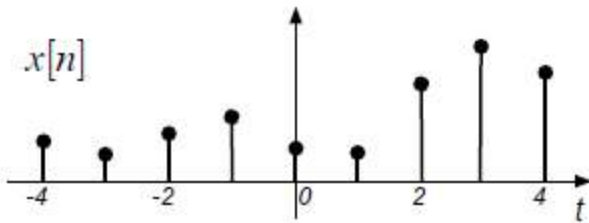
Called *upsampling*, or *interpolation*.

Time Reversal

- Continuous time: replace t with $-t$, time reversed signal is $x(-t)$



- Discrete time: replace n with $-n$, time reversed signal is $x[-n]$.

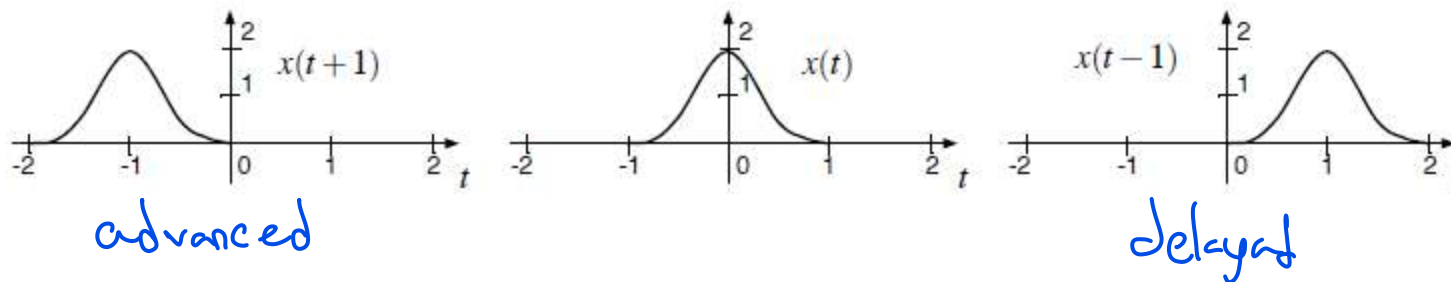


- Same as time scaling, but with $b = -1$.

Time Shift

For a continuous-time signal $x(t)$, and a time $t_1 > 0$,

- Replacing t with $t - t_1$ gives a *delayed* signal $x(t - t_1)$
- Replacing t with $t + t_1$ gives an *advanced* signal $x(t + t_1)$



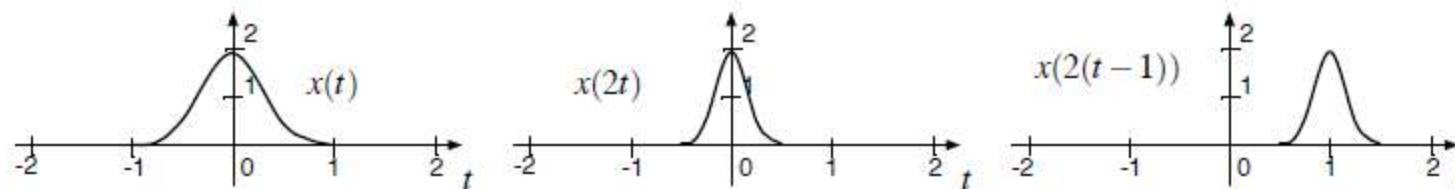
- May seem counterintuitive. Think about where $t - t_1$ is zero.

Combinations of Operations

- Time scaling, shifting, and reversal can all be combined.
- Operation can be performed in any order, but care is required.
- This *will* cause confusion.
- Example: $x(2(t-1)) \rightarrow x(2t-2)$

Scale first, then shift
Compress by 2, shift by 1

Önce 2 ötele, sonra 2 ile compress
Önce öteleme, sonra t 'nin katsayısı



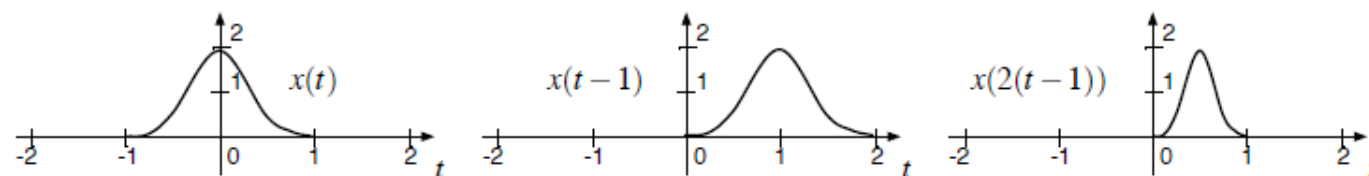
Example-1

Example $x(2(t - 1))$, continued

Shift first, then scale

Shift by 1, compress by 2

Incorrect

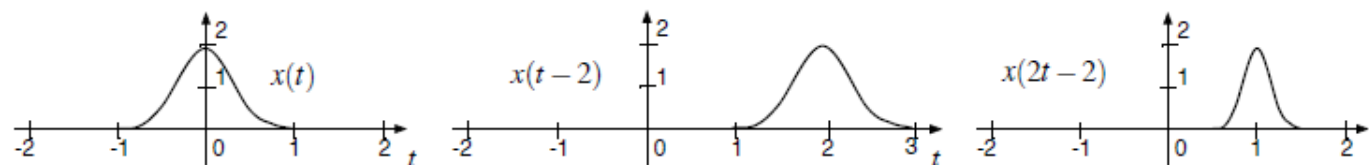


Shift first, then scale

Rewrite $x(2(t - 1)) = x(2t - 2)$

Shift by 2, scale by 2

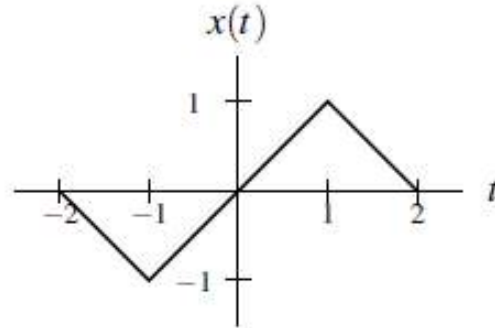
Correct



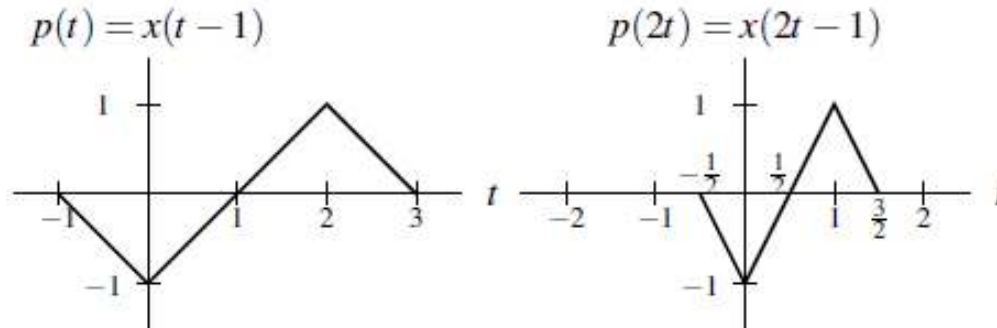
Where is $2(t - 1)$ equal to zero?

Example-2

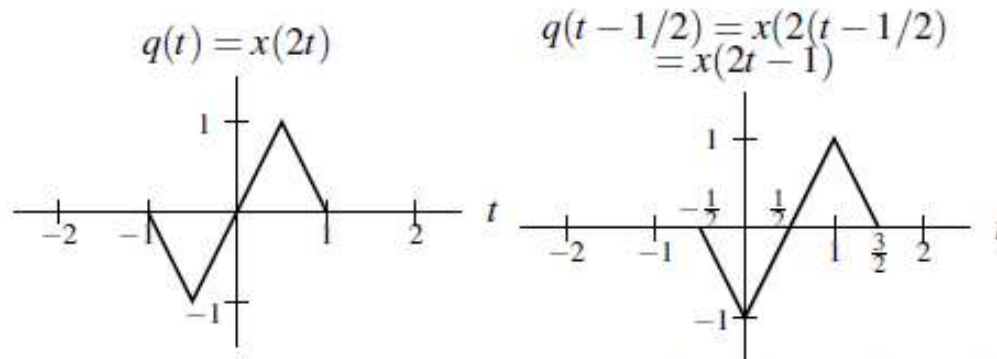
Given $x(t)$ as shown below, find $x(2t - 1)$.



time shift by 1 and then time scale by 2



time scale by 2 and then time shift by $\frac{1}{2}$



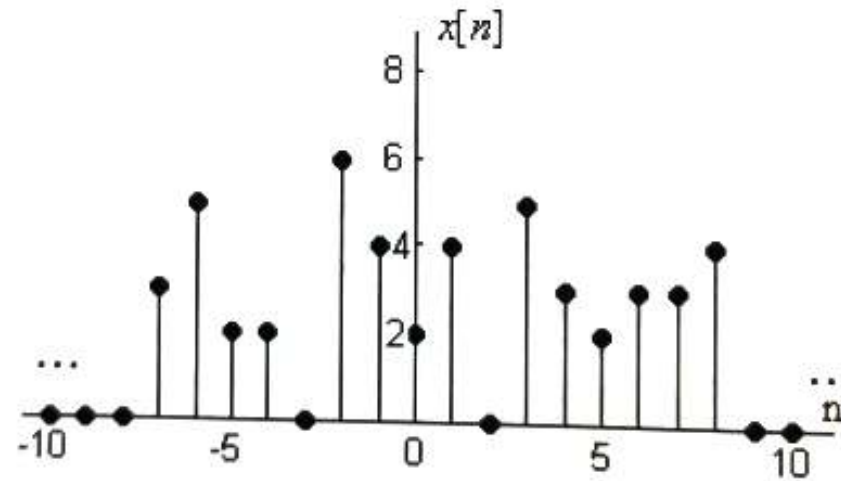
Homework (Just for you, do not e-mail)

Problemler:

2.1. $x[n]$ dizisi Şekil 2.23'de gösterilmektedir. Aşağıdaki dizileri çizin.

(a) $y[n]=x[n-3]$ (b) $y[n] = x[-n]-x[n+2]$ (c) $y[n]=x[-n+3]$ (d) $y[n]=x[6-3n]$

(e) $y[n]=x[n/2-4]$

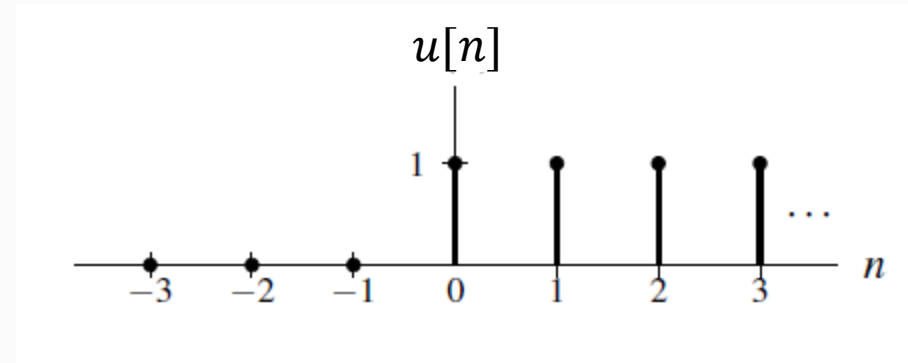


Şekil 2.23. Problem 2.1. için $x[n]$ işareti

Delta Function and Unit Step Function (Please Do Not Forget it!)

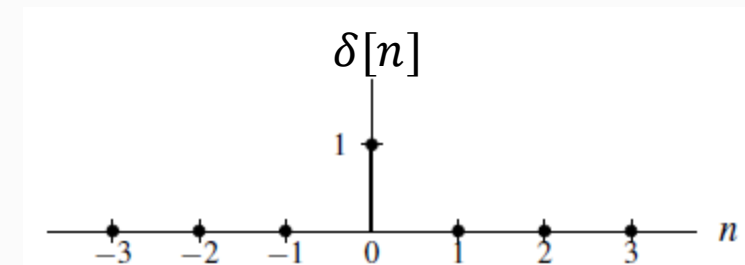
Unit Step Function

$$u(n) = \begin{cases} 1 & \text{if } n \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



Unit Impulse Function

$$\delta(n) = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

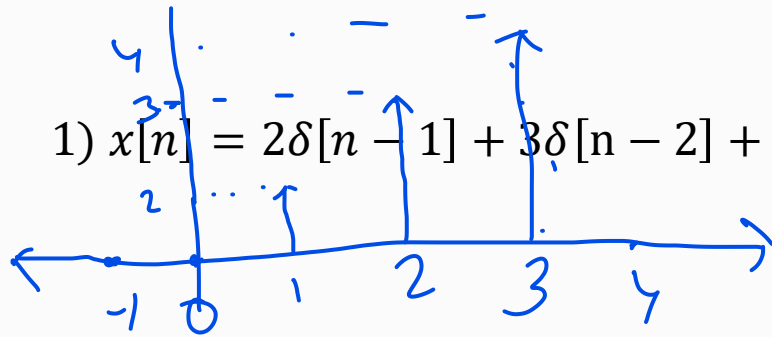


Relations

$$\delta[n] = u[n] - u[n - 1]$$

$\delta[n] = u'[n]$ dir çünkü $\frac{u[n] - u[n-1]}{1} = u[n]$ 'nin türevine eşittir.

Plot the signals given below:



1) $x[n] = 2\delta[n-1] + 3\delta[n-2] + 4\delta[n-3]$ işaretini çiziniz.

2) $y[n] = 3\delta[n-2] + u[n-5] - u[n-7]$ işaretini çiziniz.

3) $g[n] = y[n]x[n]$ ifadesini bulunuz.

4) $h[n] = y[n] + x[n]$ ifadesini bulunuz.

MATLAB kodu:

```
clc; clear all;
```

```
%%
```

```
n = [0 1 2 3 4 5 6 7];
```

```
x = [0 2 3 4 0 0 0 0];
```

```
figure(1), stem(n,x,'filled');
```

```
%%
```

```
n = [0 1 2 3 4 5 6 7];
```

```
y = [0 0 3 0 0 1 1 0];
```

```
figure(2), stem(n,y,'filled');
```

```
%%
```

```
g=y.*x;
```

```
figure(3), stem(n,g,'filled');
```

```
%%
```

```
h=y+x;
```

```
figure(4), stem(n,h,'filled');
```

*x'in n'e göre değeri-
mini 'filled' (Ciçi dolu görünüm)
ile ayık göster.*

*elementer elementer skaler
çarpma yapar.*